Coset Helly properties in an HNN extension

Human Compression project

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Let A be a group and let $\mathcal{P}$ be a family of its subgroups, closed under finite intersections. We say that $\mathcal{P}$ has the *coset Helly property* if every finite family of pairwise intersecting left cosets of members of $\mathcal{P}$ has nonempty total intersection. We include A in $\mathcal{P}$. All standard parabolics of a finite-rank flag-transitive coset geometry have this property: replace a coset of an intersection by the corresponding family of maximal-parabolic cosets, then use flag transitivity. Multiple cosets of the same maximal parabolic, if they intersect, coincide, so repeated types cause no problem.

Fix $C, D \in \mathcal{P}$ and an isomorphism $\phi : C \longrightarrow D$. Write

$$G = \langle A, t \mid t^{-1}ct = \phi(c)\ (c \in C)\rangle.$$

A subgroup $P \in \mathcal{P}$ is called compatible if $\phi(P \cap C) = P \cap D$. Put $\widehat{P} = \langle P, t\rangle$. Britton's lemma embeds the HNN extension of P along these intersections in G and gives $\widehat{P} \cap A = P$. These are retained classical normal-form facts. Unlike an arbitrary family of special subgroups, the family below keeps its local coset Helly hypothesis.

Theorem 1. *If $\mathcal{P}$ has the coset Helly property, then the family consisting of all $P \in \mathcal{P}$ and all $\widehat{P}$ for compatible $P \in \mathcal{P}$ has the coset Helly property in G. For compatible P, Q,*

$$\widehat{P} \cap \widehat{Q} = \widehat{P \cap Q}, \qquad R \cap \widehat{P} = R \cap P \quad (R \in \mathcal{P}).$$

Proof. Use the Bass–Serre tree T of this HNN extension. Its vertices are gA and its oriented edges are gC, with endpoints gA and gtA. The latter endpoint is independent of the representative because $ct = t\phi(c)$. At a vertex xA, outgoing edge directions are parametrized by left cosets of C in A, and incoming directions by left cosets of D in A.

For compatible P, the embedded Bass–Serre tree of $\widehat{P}$ has vertices hA and edges hC, $h \in \widehat{P}$. Its injectivity follows from the same reduced word theorem, since the associated intersections are exactly $P \cap C$ and $P \cap D$. Denote this connected subtree by T_P. For a base subgroup $R \leq A$ instead use the singleton subtree $\{A\}$. Attach to a coset $g\widehat{P}$ the subtree gT_P and to a coset gR the vertex gA. These assignments are independent of the chosen coset representative.

If a vertex $v = xA$ belongs to such a support subtree, the corresponding coset meets xA. After multiplication by x^{-1} this intersection is a left coset in A of P, or of R in the base-subgroup case. For example, choose $y \in g\widehat{P} \cap xA$; the intersection is $y(\widehat{P} \cap A) = yP$. An incident edge belongs to gT_P precisely when this local P coset meets the local C or D coset describing the direction of that edge. This follows directly from the vertex and edge descriptions of the embedded tree.

We need one transport observation. Suppose two global cosets meet, and a vertex v lies in both support subtrees. Choose a common element z of the cosets. Both support subtrees contain zA, so they contain the whole finite path from zA to v. Start with the common representative z at its first vertex. At the next edge, the two local subgroup cosets intersect each other and each meets the local associated-subgroup coset of that edge. Local coset Helly supplies a common element of all three. For an outgoing edge multiply this representative on the right by t; for an incoming edge multiply it by t^{-1}. The result lies at the next vertex and still belongs to both global cosets, because both subgroups

contain t. If either subgroup was a base subgroup, its support was a singleton and the path has length zero, so no multiplication is needed. Repeating the step gives a common representative in xA.

Now take finitely many pairwise intersecting global cosets. Their support subtrees intersect pairwise. The elementary Helly property for subtrees gives a common vertex $v = xA$. By the transport observation every pair of local fibers at v intersects. These fibers are cosets of members of $\mathcal{P}$, so local coset Helly supplies an element in all of them. Multiplying by x proves the first assertion.

For the intersection formula, first observe that $P \cap Q$ is compatible: an isomorphism commutes with intersections. Thus $K = \widehat{P \cap Q}$ is contained in both groups. Take $g \in \widehat{P} \cap \widehat{Q}$ and follow the finite path from A to gA. Both embedded subtrees contain it. We show inductively that every vertex on this path is kA for some $k \in K$. This holds initially with $k = 1$. Translate a current vertex kA and its next edge by k^{-1}. If that edge is outgoing, its local coset aC meets both P and Q; if incoming, its local coset aD does. Since $P \cap Q$ contains 1, the three local cosets meet pairwise. The local Helly property supplies $p \in P \cap Q$ in the relevant coset. The next vertex is therefore $kptA$ or $kpt^{-1}A$, respectively, and its representative is again in K. At the end $gA = kA$, and hence $k^{-1}g \in A \cap \widehat{P} \cap \widehat{Q} = P \cap Q \leq K$. This proves the first equality. The second follows immediately from $\widehat{P} \cap A = P$. □

Corollary 2. *Let $(A, (A_i)_{i\in I})$ be a flag-transitive coset geometry of finite rank. Let C, D be standard parabolics, and let $\phi : C \to D$ be an isomorphism. Choose a partition $\mathcal{S}$ of I such that every $A_s = \bigcap_{i\in s} A_i$, $s \in \mathcal{S}$, is compatible, and choose $F \subseteq I$. In the HNN extension take maximal parabolics*

$$G_s = \langle A_s, t\rangle \quad (s \in \mathcal{S}), \qquad G_* = A_F.$$

This is a flag-transitive coset geometry. For a set J of its types, write $E = \bigcup_{s\in J\cap\mathcal{S}} s$. Then

$$G_J = \begin{cases} \langle A_E, t\rangle, & * \notin J, \\ A_{E\cup F}, & * \in J. \end{cases}$$

Its chamber stabilizer is A_I.

Proof. Take $\mathcal{P}$ to be all standard parabolics of the input, including A. These have the local coset Helly property explained above. Theorem 1 applies to all the displayed maximal parabolics. A flag is a finite family of pairwise intersecting such cosets, so it has a common representative. It is therefore a translate of a standard flag and extends to a standard chamber. The intersection formulas follow by induction from the theorem, with the usual empty-intersection convention. For the complete type set $E = I$, so the Borel is A_I. □

Corollary 3. *Suppose the input in Corollary 2 is residually connected. Let $C = A_J$, $D = A_K$, and suppose a specified bijection $f : I \setminus J \longrightarrow I \setminus K$ satisfies*

$$\phi(A_{J\cup X}) = A_{K\cup f(X)} \qquad (X \subseteq I \setminus J).$$

Take $\mathcal{S}$ to be the connected classes of the arrows $i \to f(i)$ on I, and put $F = J \setminus K$. Then the resulting HNN geometry is residually connected. Its $*$ panels are infinite. Its panel of type $s \in \mathcal{S}$ has size

$$[A_{I\setminus(s\setminus F)} : A_I].$$

In particular it is firm when the input is firm.

Proof. Write $H = A_I$ and $R_i = A_{I\setminus\{i\}}$. The displayed compatibility gives $\phi(H) = H$ and $\phi(R_i) = R_{f(i)}$ for $i \notin J$. Each class s satisfies $f(s \setminus J) = s \setminus K$. Consequently $\phi(A_s \cap C) = A_s \cap D$, so the previous corollary applies. Its minimal parabolics are

$$M_s = A_{I\setminus(s\setminus F)}, \qquad M_* = \langle H, t\rangle.$$

Residual connectivity of the input gives the elementary generation rule

$$A_{I\setminus E} = \langle H, R_i \ (i \in E)\rangle \quad (E \subseteq I).$$

For $E = \varnothing$ this just retains the Borel H; for nonempty E it follows by induction from generation by maximal parabolics in each connected residue of rank at least two. This is the Borel-preserving version of the usual residual generation criterion.

Let W be a subset of the new type set $\mathcal{S} \cup \{*\}$, and put $E = \bigcup_{s\in W\cap\mathcal{S}} s$. If $* \notin W$, then

$$\langle H, M_s \ (s \in W)\rangle = A_{I\setminus(E\setminus F)} = G_{(\mathcal{S}\cup\{*\})\setminus W}.$$

If $* \in W$, this generated group also contains t. Every $i \in E\cap F$ is in the range but not the domain of f. It therefore has a predecessor $j \in E \setminus F$ with $f(j) = i$. Conjugation by t brings R_i into the group from R_j. Thus in this case the generated group is $\langle H, t, R_i \ (i \in E)\rangle = \langle A_{I\setminus E}, t\rangle$, again exactly $G_{(\mathcal{S}\cup\{*\})\setminus W}$.

We have proved the same generation rule for every new standard parabolic. If at least two types remain, each of its minimal generators belongs to one of the parabolics obtained by deleting a different remaining type. Those parabolics therefore generate it. The residual generation criterion proves residual connectivity.

Since $\phi(H) = H$, the subgroup $\langle H, t\rangle$ is an extension of H by the infinite cyclic group generated by the stable letter. Its index over H is infinite. The other panel indices are the stated indices of M_s over H. Every class has a point outside F: a point in F has a predecessor outside F. If the input is firm, the group M_s therefore contains some R_i strictly larger than H. All these panels have at least two elements, as required. □

The partition and compatibility conditions in Corollary 2 are explicit. No isomorphism of unlabelled subgroup lattices is silently promoted to an action on named types. If a specified partial bijection $I\setminus J \to I\setminus K$ describes an isomorphism $A_J \to A_K$ on the indexed parabolics, its connected type classes give the partition used in Piedade–Tranchida, arXiv:2505.24662v1, Section 3.3. Verifying this precise condition recovers their forward flag-transitivity and intersection assertions. Corollary 3 supplies residual connectivity, firmness and exact panel sizes under the explicit indexed compatibility.

The proof combines classical Bass–Serre normal forms with the elementary Helly property for subtrees and local coset Helly. The transfer argument is written here to repair the use of the unrestricted special-subgroup intersection lemma, which fails in the separate obstruction note. Historical priority for this transfer formulation has not been assessed. This manuscript is a private draft awaiting distinct review; no numerical or complete-original credit is assigned.